

Mohammed Hassan

Efficient Embodied AI and Human-Controllable Generative Systems

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EDUCATION

The University of Texas at Austin

M.S. in Computer Science

Expected May 2027

B.S. in Computer Science and B.S. in Mathematics

May 2025

RESEARCH EXPERIENCE

Georgia Institute of Technology | EIC Lab

Jul 2025 - Present

Research Intern

Supervision: Prof. Yingyan (Celine) Lin; Prof. Chaojian Li (T3R)

T3R | [Project page](#)

First author | ASP-DAC 2027, accepted

Training-Free Two-Stage Token Refinement Towards Efficient and Robust VLA Models

- Developed a training-free VLA inference framework combining segmentation-guided visual-token pruning with saliency-guided attention refinement.
- Removed **85% of scene-camera tokens** on LIBERO-Spatial while achieving **96% task success** (97% unpruned); improved success by up to **7 percentage points** under object-placement shifts.
- Evaluated OpenVLA-OFT, CogACT and pi0.5 across LIBERO, SIMPLER and RoboTwin to characterize pruning performance in single- and multi-camera settings.

MIRA-Quant

Co-author

- Contributed to a multi-agent iterative-reasoning system for automated LLM quantization. Matched expert-tuned NVIDIA Model Optimizer references within **0.3 perplexity** across 15 configurations spanning five LLMs and three post-training quantization methods.

The University of Texas at Austin | VITA Lab

Research Intern

Supervision: Prof. Atlas Wang and Dr. Hezhen (Alex) Hu

MAESTRO | [Project page](#)

First author | CHI 2027, under review

Agentic Long-Form Music-to-3D Dance Composition and Editing

- Built a section-aware dance-authoring framework that coordinates LODGE and EDGE through language-based planning, motion execution and measured evaluation.
- Implemented interval-scoped language editing, feedback-guided refinement and branching checkpoints so users can revise local motion, compare alternatives and restore accepted choreography.
- Evaluated the system with **12 dance-experienced participants**. MAESTRO received **47 of 72 overall-preference selections (65.3%)** in blind comparisons with LODGE and EDGE.

ADDITIONAL RESEARCH

Improving BB84 Efficiency with Delayed Measurement via Quantum Memory

QIP 2025, Raleigh, NC | Research: Aug - Oct 2024 | [arXiv:2410.21191](#)

- Proposed quantum-memory-assisted delayed measurement to eliminate basis-mismatch discards in an idealized BB84 analysis; examined authentication, coherence and timing requirements.

Greenify: Energy-Aware Neural Architecture Search

Jan - Apr 2025

Preprint

- Built a differentiable NAS framework with PyTorch and CodeCarbon; reduced measured inference energy by **18% on MNIST** with less than 0.2% accuracy loss.

Hyperparameter Optimization of Diffusion Models

Oct - Dec 2024

Preprint

- Compared random/grid search, Bayesian optimization and gradient-based methods, examining their suitability for flow-based generative models.

PROFESSIONAL AND TEACHING EXPERIENCE

Microsoft | Software Engineer

Aug 2025 - Present

- Own backend services and REST APIs for DNS-based threat intelligence, with service-level objectives for availability, latency and error rate.
- Lead infrastructure changes and design reviews for staged Azure DevOps/EV2 rollouts, including canary deployment and automatic rollback.

Microsoft | Software Engineering Intern

Jun - Aug 2024

- Built a serverless Azure Function App and ML model to detect potential data exfiltration from virtual-network DNS queries, including feature engineering, training and evaluation.

Google | STEP Intern

May - Aug 2023

- Developed a Java/Flume data pipeline and integrated it with notification systems to support targeted campaigns.

UT Austin | Teaching Assistant, CS 311 Discrete Math

Aug 2022 - May 2025

- Mentored **400+ students** in logic, combinatorics, graph theory and algorithm verification.

RELEVANT COURSEWORK

Graduate: Reinforcement Learning (A), Deep Learning (A), Deep Generative Models (A).

In progress: Optimization; Planning, Search, and Reasoning Under Uncertainty.

Foundations: Linear Algebra, Probability and Statistics, Algorithms and Complexity, Geometric Foundations of Data Science, Energy Efficient Computing.

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, C++, Java, SQL, shell scripting.

ML and research: PyTorch, TensorFlow/Keras, JAX, Hugging Face Transformers, scikit-learn, Optuna, CodeCarbon, LIBERO, Matplotlib, TensorBoard.

Platforms: Git, Linux/UNIX, Azure, Google Cloud Platform, Azure DevOps, Docker, Kubernetes.

Languages: Arabic and English (native/bilingual).